

Problem 1.1 Properties of convolution

Recall that 1D convolution between two signals $f, g \in \mathbb{R}^N$ is defined:

$$(f * g)[n] = \sum_{k=0}^{N-1} f[n-k]g[k]. \quad (1)$$

(a) Construct a matrix multiplication that produces the same result as a 1D convolution with a given filter. In other words, given a filter f , construct a matrix H such that $f * g = Hg$ for any input g . Here Hg denotes matrix multiplication between the matrix H and vector g .

(2 points)

Note: One option is to define H in “bracket” notation using “...” symbols, e.g., $H = \begin{bmatrix} 0 & 1 & \dots & N-1 \\ N-1 & N-2 & \dots & 1 \end{bmatrix}$. Another option is to explicitly define the entries H_{ij} .

$$\begin{aligned} \text{a) } (f * g) &= \begin{bmatrix} f[0]g[0] & f[1]g[0] + f[0]g[1] & \dots & f[N-1]g[0] + f[N-2]g[1] + \dots + f[0]g[N-1] \end{bmatrix} = Hg \\ \Rightarrow [Hg]_i &= \sum_{j=0}^{N-1} H_{ij} g_j, \quad H_{ij} = \begin{cases} f[i-j], & i \geq j \\ 0, & i < j \end{cases} \end{aligned}$$

(b) In class, we showed that convolution with a 2D Gaussian filter can be performed efficiently as a sequence of convolutions with 1D Gaussian filters. This idea also works with other kinds of filters. We say that a 2D filter $F \in \mathbb{R}^{N \times N}$ is separable if $F = uv^T$ for some $u, v \in \mathbb{R}^N$, i.e., F is the *outer product* of u and v . Show that if F is separable, then the 2D convolution $G * F$ can be computed as a sequence of two one-dimensional convolutions **(2 points)**.

$$\begin{aligned} \text{b) Given } F &= uv^T \Rightarrow F[m, n] = u[m] \cdot v[n] \\ (G * F)[i, j] &= \sum_{m=0}^{N-1} \sum_{n=0}^{N-1} G[i-m, j-n] F[m, n] \\ &= \sum_{m=0}^{N-1} \sum_{n=0}^{N-1} G[i-m, j-n] u[m] v[n] \\ &= \sum_{m=0}^{N-1} u[m] \left(\sum_{n=0}^{N-1} G[i-m, j-n] v[n] \right) \\ \text{let } G'[i, j] &= (G * v^T)[i, j] = \sum_{n=0}^{N-1} G[i, j-n] v[n] = G * v^T \\ (G * F)[i, j] &= \sum_{m=0}^{N-1} G'[i-m, j] u[m] = G' * u = G * v^T * u \end{aligned}$$